Part I

Euclid



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Little is known about the person of Euclid (Εὐκλείδης, c. 320–c. 275 BC). Proclus (410–485 AD) edited and commented his famous work, the *Elements* (στοιχεῖα), and summed up the meager knowledge about Euclid: *Euclid put together the* Elements*, collected many of Eudoxus' theorems, perfected many of Theaitetus', and also brought to irrefragable demonstration the things which were only somewhat loosely proved by his predecessors. This man lived in the time of the first Ptolemy. For Archimedes, who came immediately after the first (Ptolemy), makes mention of Euclid: and, further, they say that Ptolemy once asked him if there was in geometry any shorter way than that of the elements, and he answered that there was no royal road to geometry. He is then younger than the pupils of Plato but older than Eratosthenes and Archimedes; for the latter were contemporary with one another, as Eratosthenes somewhere says* (translation[1] from Heath 1925).

By interpolating between Plato's (427–347 BC) students, Archimedes (287–212 BC), and Eratosthenes (c. 275–195 BC), we can guess that Euclid lived around 300 BC; Ptolemy reigned 306–283. It is likely that Euclid learned mathematics in Athens from Plato's students. Later, he founded a school at Alexandria, and this is presumably where most of the *Elements* was written. An anecdote relates how "some one who had begun to read geometry with Euclid, when he had learned the first theorem, asked Euclid, 'But what shall I get by learning these things?' Euclid called his slave and said 'Give him threepence, since he must make gain out of what he learns.'"

In later parts of this book, we will present two giants of mathematics—Newton and Gauß—and two great mathematicians—Fermat and Hilbert. Archimedes is the third giant, in our subjective reckoning. Euclid's ticket to enter our little Hall of Fame is not a wealth of original ideas, but—besides his algorithm for the gcd—his insistence that everything must be proven from a few axioms, as he does in his systematic collection and orderly presentation of the mathematical thought of his times in the thirteen books (chapters) of his *Elements*. Probably written around 300 BC, he presents his material in the axiom-definition-lemma-theorem-proof style which—somewhat modified—has survived through the millenia. Euclid's methods go back to Aristotle's (384–322 BC) peripatetic school, and to the Eleatics.

After the Bible, the *Elements* is apparently the most often printed book, an eternal bestseller with a vastly longer half-life (before oblivion) than current-day bestsellers. It was *the* mathematical textbook for over two millenia. (In spite of their best efforts, the authors of the present text expect it to be superseded long before 4298 AD.) Even in 1908, a translator of the *Elements* exulted: *Euclid's*

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work will live long after all the text-books of the present day are superseded and forgotten. It is one of the noblest monuments of antiquity; no mathematician worthy of the name can afford not to know Euclid. Since the invention of non-Euclidean geometry and the new ideas of Klein and Hilbert in the 19th century, we don't take the *Elements* quite that seriously any longer.

In the Dark Ages, Europe's intellectuals were more interested in the maximal number of angels able to dance on a needle tip, and the *Elements* mainly survived in the Arabic civilization. The first translation from the Greek was done by Al-Ḥajjāj bin Yūsuf bin Maṭar (c. 786–835) for Caliph Hārūn al-Rashīd (766–809). These were later translated into Latin, and Erhard Ratdolt produced in Venice the first printed edition of the *Elements* in 1482; in fact, this was the first mathematics book to be printed. On page 23 we reproduce its first page from a copy in the library of the University of Basel; the underlining is possibly by the lawyer Bonifatius Amerbach, its 16th century owner, who was a friend of Erasmus.



Most of the *Elements* deals with geometry, but Books 7, 8, and 9 treat arithmetic. Proposition 2 of Book 7 asks: "Given two numbers not prime to one another, to find their greatest common measure", and the core of the algorithm goes as follows: "Let AB, CD be the two given numbers not prime to one another [...] if CD does not measure AB, then, the lesser of the numbers AB, CD being continually subtracted from the greater, some number will be left which will measure the one before it" (translation from Heath 1925).

Numbers here are represented by line segments, and the proof that the last number left (dividing the one before it) is a common divisor and the greatest one is carried out for the case of two division steps ($\ell = 2$ in Algorithm 3.6). This is Euclid's algorithm, "the oldest nontrivial algorithm that has survived to the present day" (Knuth 1998, §4.5.2), and to whose

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understanding the first part of this text is devoted. In contrast to the modern version, Euclid does repeated subtraction instead of division with remainder. Since some quotient might be large, this does not give a polynomial-time algorithm, but the simple idea of removing powers of 2 whenever possible already achieves this (Exercise 3.25).

In the geometric Book 10, Euclid repeats this argument in Proposition 3 for "commensurable magnitudes", which are real numbers whose quotient is rational, and Proposition 2 states that if this process does not terminate, then the two magnitudes are incommensurable.

The other arithmetical highlight is Proposition 20 of Book 9: "Prime numbers are more than any assigned multitude of prime numbers." Hardy (1940) calls its proof "as fresh and significant as when it was discovered—two thousand years have not written a wrinkle on [it]". (For lack of notation, Euclid only illustrates his proof idea by showing how to find from three given primes a fourth one.)

It is amusing to see how after such a profound discovery comes the platitude of Proposition 21: "If as many even numbers as we please be added together, the whole is even." The *Elements* is full of such surprises, unnecessary case distinctions, and virtual repetitions. This is, to a certain extent, due to a lack of good notation. Indices came into use only in the early 19th century; a system designed by Leibniz in the 17th century did not become popular.

Euclid authored some other books, but they never hit the bestseller list, and some are forever lost.

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Die ganzen Zahlen hat der liebe Gott gemacht,
alles andere ist Menschenwerk.[1]

Leopold Kronecker (1886)

"I only took the regular course." "What was that?" enquired Alice.
"Reeling and Writhing, of course, to begin with," the Mock Turtle
replied: "and then the different branches of Arithmetic—Ambition,
Distraction, Uglification, and Derision."

Lewis Carroll (1865)

Computation is either perform'd by *Numbers*, as in Vulgar
Arithmetick, or by *Species* [variables][2], as usual among Algebraists.
They are both built on the same Foundations, and aim at the same End,
viz. Arithmetick Definitely and Particularly, *Algebra* Indefinitely and
Universally; so that almost all Expressions that are found out by this
Computation, and particularly Conclusions, may be called *Theorems*.
[...] Yet Arithmetick in all its Operations is so subservient to Algebra,
as that they seem both but to make one perfect *Science of Computing*.

Isaac Newton (1728)

It is preferable to regard the computer as a handy device
for manipulating and displaying symbols.

Stanisław Marcin Ulam (1964)

In summo apud illos honore geometria fuit, itaque nihil
mathematicis inlustrius; at nos metiendi ratiocinandique
utilitate huius artis terminauimus modum.[3]

Marcus Tullius Cicero (45 BC)

But the rest, having no such grounds [religious devotion] of hope,
fell to another pastime, that of computation.

Robert Louis Stevenson (1889)

Check your math and the amounts entered
to make sure they are correct.

State of California (1996)

[1] God created the integers, all else is the work of man.

[2] [Text in brackets added by the authors, also in other quotations.]

[3] Among them [the Greeks] geometry was held in highest esteem, nothing was more glorious than mathematics; but we have restricted this science to the practical purposes of measuring and calculating.

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